Ratan familiarizing himself with the processes associated with the coke ovens and the furnaces, the hottest places in the steel plant. Their temperature rises to over 1,000 degrees Celsius, making it unbearably hot to stand in the vicinity. Some told him to ‘shovel limestone into the open-hearth furnaces’. He reminisces that another day somebody would say, ‘you need to know how to tell temperature from the colour of the flames so I’ll teach that to you’ and ‘so go on top of the coke ovens’.

These were not easy orders to comply with in the roasting dry heat. When one stood on top of the coke oven, ‘your shoes’ soles would melt, your watch strap and your pen in your pocket’ burned your skin, Ratan recalls. But in the process, he learned a lot about steelmaking and the various processes it entailed.

After two years, Ratan was moved to the engineering office. Here he spent one year as the technical assistant to S.K. Nanavati, the director-in-charge at the steel plant. Work began to get more exciting, especially when new projects were taken up. The expansion projects he was assigned to were quite interesting, he remembers.

Ratan was ‘glued to work’, says K.P. Mahalingam, chief engineer at the steel giant at the time, who was mentor to the young Tata for three years. He comments that he had found Ratan to be ‘shy by nature’, but very polite with everybody.

One of the areas that caught Ratan’s attention was the ‘structural shops’, a kind of workshop established to undertake the structural maintenance of the plant and some limited works in the engineering divisions. Ratan recommended the transformation and expansion of these shops. He redesigned and mechanized them to undertake repairing and make new products.

Ratan’s innovative ideas helped immensely in changing the thought process of the steel company employees. Most importantly, he made the staff realize that there was room for rethinking their ways of running the steel plant, says N.P. Sinha, the former vice-president of TISCO.

The changes he brought about in the structural shops also had other benefits. Prior to the liberalization of the economy in 1991, foreign exchange was scarce and there were severe restrictions on the import of capital equipment, though they were critically required as replacements and for modernization. The cost involved was high and the process